Chapter 4

Rotations & Angular Momentum

4.1 Rotations in 3D Space

Rotations play a central role in quantum mechanics and serve as a gentle introduction to the *group-theoretic framework* that underlies modern physics. Group theory provides the mathematical language for describing symmetries in a precise way. Informally, a group is a set of transformations that can be combined and inverted, capturing the structure of symmetry operations. By studying the group of rotations, we will uncover an underlying algebraic structure that allows us to derive quantum states systematically, understand their behavior under rotations, and describe phenomena such as spin.

We begin by studying the group of *proper rotations*, denoted SO(3). Given a unit vector $\hat{\mathbf{u}} = (u_x, u_y, u_z)$, a proper rotation R is a linear map from $\hat{\mathbf{u}}$ to another unit vector

$$\hat{\mathbf{u}}' = R\hat{\mathbf{u}}$$

and it preserves the orientation of a right-handed triplet. Namely,

$$R\hat{\mathbf{z}} \cdot (R\hat{\mathbf{x}} \times R\hat{\mathbf{y}}) = \det(R) \det \begin{bmatrix} \hat{\mathbf{z}} & \hat{\mathbf{x}} & \hat{\mathbf{y}} \end{bmatrix} = \det(R) = 1$$

Since R is a linear map between unit vectors, it must be orthogonal:

$$\hat{\mathbf{u}}' \cdot \hat{\mathbf{u}}' = (R\hat{\mathbf{u}}) \cdot (R\hat{\mathbf{u}}) = \hat{\mathbf{u}}^T R^T R \hat{\mathbf{u}} = 1 \quad \Rightarrow \quad R^T R = \mathbb{1}$$

Hence, the elements of SO(3) are represented by matrices with the following properties:

$$R^T R = R R^T = \mathbb{1} \quad \text{and} \quad \det(R) = 1 \tag{4.1}$$

4.1.1 Rotations About a Fixed Axis

Consider a rotation about some axis $\hat{\mathbf{n}}$ by an angle θ in which the sense is determined by the right hand rule. Denote this rotation by $R(\hat{\mathbf{n}}, \theta)$. Rotations about the same axis commute and the angles

add under matrix multiplication:

$$R(\hat{\mathbf{n}}, \theta_1)R(\hat{\mathbf{n}}, \theta_2) = R(\hat{\mathbf{n}}, \theta_2)R(\hat{\mathbf{n}}, \theta_1) = R(\hat{\mathbf{n}}, \theta_1 + \theta_2)$$

Consecutive rotations about different axes do not commute; the group of rotations is called "non-abelian" for this reason. The rotations about the three coordinate axes are

$$R(\hat{\mathbf{x}}, \theta) = R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \tag{4.2}$$

$$R(\hat{\mathbf{y}}, \theta) = R_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \tag{4.3}$$

$$R(\hat{\mathbf{z}}, \theta) = R_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.4}$$

and you can work out for yourself simple examples where $R_z(\theta_1)R_x(\theta_2) \neq R_z(\theta_2)R_x(\theta_1)$ for instance. I'll probably ask you to perform rotations on your phone in class...

One can show that any proper rotation can be represented as $R(\hat{\mathbf{n}}, \theta)$, for some axis $\hat{\mathbf{n}}$ and some angle θ. Thus, there is no loss of generality in writing a proper rotation in this form. We will call this the *axis-angle parameterization* of the rotations.

4.1.2 Infinitesimal Rotations

Consider an infinitesimal rotation of the vector $\mathbf{u}$ about the axis $\hat{\mathbf{n}}$ by a small angle $\delta\theta$ (see Figure 4.1 for a geometric argument):

$$R(\hat{\mathbf{n}}, \delta\theta)\mathbf{u} = \mathbf{u} + \delta\theta\hat{\mathbf{n}} \times \mathbf{u} + \mathcal{O}(\delta\theta^2) \tag{4.5}$$

The cross product can be written as a matrix equation using the following identity:

$$\mathbf{a} \times \mathbf{u} = A\mathbf{u} = \begin{pmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{pmatrix} \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix} \tag{4.6}$$

The matrix A is antisymmetric ($A^T = -A$) and admits the following decomposition:

$$A = \mathbf{a} \cdot \boldsymbol{\mathcal{J}} = a_x\mathcal{J}_x + a_y\mathcal{J}_y + a_z\mathcal{J}_z \tag{4.7}$$

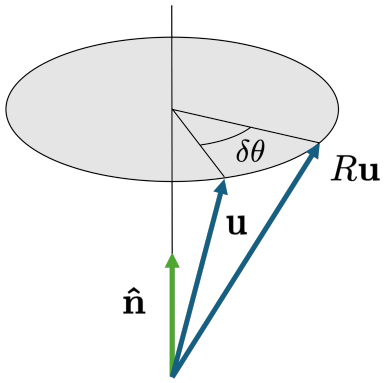


Figure 4.1: Infinitesimal rotation of the vector $\mathbf{u}$ about the axis specified by $\hat{\mathbf{n}}$. The component of $\mathbf{u}$ parallel to $\hat{\mathbf{n}}$ remains constant while the perpendicular component rotates in a positive sense (given by the right hand rule) around the axis defined by $\hat{\mathbf{n}}$.

where $\boldsymbol{\mathcal{J}} = (\mathcal{J}_x, \mathcal{J}_y, \mathcal{J}_z)$ is a "vector" of the matrices

$$\mathcal{J}_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathcal{J}_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad \mathcal{J}_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{4.8}$$

The matrices $\mathcal{J}_x, \mathcal{J}_y$ and $\mathcal{J}_z$ encode the geometric structure of rotations (and ultimately the structure of 3D Euclidean space). The non-commutativity of rotations is fully captured by the commutators, which you can verify by direct calculation:

$$[\mathcal{J}_x, \mathcal{J}_y] = \mathcal{J}_z, \quad [\mathcal{J}_z, \mathcal{J}_x] = \mathcal{J}_y, \quad [\mathcal{J}_y, \mathcal{J}_z] = \mathcal{J}_x \tag{4.9}$$

Any infinitesimal rotation can be written as

$$R(\hat{\mathbf{n}}, \delta\theta) = 1 + \delta\theta \hat{\mathbf{n}} \cdot \boldsymbol{\mathcal{J}} \tag{4.10}$$

and it follows that any finite rotation can then be written as the matrix exponential

$$R(\hat{\mathbf{n}}, \theta) = \exp\left(\theta \hat{\mathbf{n}} \cdot \boldsymbol{\mathcal{J}}\right) \tag{4.11}$$

It turns out that knowledge of the commutation relations in Eq. (4.9) is enough to compute the multiplication of any two finite rotations in terms of axis-angle parameters.

4.2 Rotations in Quantum Mechanics

Our goal for this section is to find operators that act on quantum states and represent rotations at the classical level–rotations in 3D Euclidean space. We saw in chapter #3 there was a sense in which a spin-1/2 system precesses in a magnetic field. Specifically, the expectation value of the

spin vector rotates over time around the direction of the magnetic field. In the context of spin precession, a unitary operator evolves the state over time in direct correspondence with a rotating classical vector **S**. We postulate for each proper rotation matrix R in 3D Euclidean space, there corresponds a unitary operator $U(R)$ acting on state vectors. The specific form of $U(R)$ depends on the system, but the essential properties of quantum rotations follow from the properties of classical rotations.

4.2.1 Angular Momentum

Recall that Eq. (4.11) for a classical rotation depends on the product $\theta\hat{\mathbf{n}}$. Define $\boldsymbol{\theta} = \theta\hat{\mathbf{n}}$ such that for an infinitesimal rotation

$$R(\boldsymbol{\theta}) = \mathbb{1} + \boldsymbol{\theta} \cdot \boldsymbol{\mathcal{J}} + \cdots \tag{4.12}$$

which implies

$$\mathcal{J}_x = \left.\frac{\partial R(\boldsymbol{\theta})}{\partial \theta_x}\right|_{\boldsymbol{\theta}=0}, \quad \mathcal{J}_y = \left.\frac{\partial R(\boldsymbol{\theta})}{\partial \theta_y}\right|_{\boldsymbol{\theta}=0}, \quad \mathcal{J}_z = \left.\frac{\partial R(\boldsymbol{\theta})}{\partial \theta_z}\right|_{\boldsymbol{\theta}=0} \tag{4.13}$$

The unitary operator that corresponds to $R(\boldsymbol{\theta})$ is ultimately a function of $\boldsymbol{\theta}$, and for an infinitesimal rotation, we can approximate $U(\boldsymbol{\theta})$ by the leading terms of its Taylor series

$$U(\boldsymbol{\theta}) = \mathbb{1} - \frac{i}{\hbar}\boldsymbol{\theta} \cdot \mathbf{J} + \cdots \tag{4.14}$$

where (by analogy with the classical case)

$$J_x = i\hbar \left.\frac{\partial U(\boldsymbol{\theta})}{\partial \theta_x}\right|_{\boldsymbol{\theta}=0}, \quad J_y = i\hbar \left.\frac{\partial U(\boldsymbol{\theta})}{\partial \theta_y}\right|_{\boldsymbol{\theta}=0}, \quad J_z = i\hbar \left.\frac{\partial U(\boldsymbol{\theta})}{\partial \theta_z}\right|_{\boldsymbol{\theta}=0} \tag{4.15}$$

Formally, components of **J** are the *generators of rotation* in quantum mechanics. When a classical system is invariant under rotation about some axis, the angular momentum about that axis is conserved. Likewise, when the Hamiltonian for a quantum system commutes with the generator of rotation about some axis, the expectation value of the generator is conserved. We define the **angular momentum** of a quantum system as the vector operator $\mathbf{J} = (J_x, J_y, J_z)$. The factor $\hbar$ ensures (i) **J** has the units of angular momentum and (ii) **J** corresponds to classical angular momentum in the appropriate limit. The factor of i ensures the generators are self-adjoint, i.e. $J_x^\dagger = J_x$, etc. Any finite rotation corresponds to the unitary operator

$$U(\boldsymbol{\theta}) = \exp\left(-\frac{i}{\hbar}\boldsymbol{\theta} \cdot \mathbf{J}\right) = \exp\left(-\frac{i}{\hbar}\theta\hat{\mathbf{n}} \cdot \mathbf{J}\right) \tag{4.16}$$

4.2.2 The Adjoint Formula

Up to this point, we have constructed quantum rotation operators by analogy with classical rotations. We began with the geometric structure of SO(3), extracted its generators $\boldsymbol{\mathcal{J}}$, and then introduced self-adjoint operators **J** whose exponentials generate unitary transformations on Hilbert

space. We have not, however, explicitly related $R \to U(R)$. The assignment must be compatible with how rotations act on ordinary vectors.

Consider an operator-valued vector, or vector operator, denoted $\mathbf{A} = (A_x, A_y, A_z)$, where each component represents a Hermitian operator ($A_i^\dagger = A_i$). In other words, $\mathbf{A}$ is a vector whose components are observables. If $\mathbf{A}$ was an ordinary vector in 3D Euclidean space, then a rotation would be a linear mapping $\mathbf{A} \to R\mathbf{A}$. Now consider a unitary transformation $U(R)$ corresponding to the 'classical rotation' R. On a quantum system in the state $|\psi\rangle$,

$$\left|\psi'\right\rangle = U(R)\left|\psi\right\rangle \tag{4.17}$$

If $U(R)$ truly represents the classical rotation R, then the expectation values must transform like ordinary vectors. More precisely, when the quantum state is rotated, the expectation value of the vector operator rotates exactly like a classical vector in $\mathbb{R}^3$. Mathematically,

$$\left\langle\psi'\right| \mathbf{A} \left|\psi'\right\rangle = R\left\langle\psi\right| \mathbf{A} \left|\psi\right\rangle, \tag{4.18}$$

Since $|\psi'\rangle = U(R)|\psi\rangle$, it follows that

$$\left\langle\psi\right| U^\dagger \mathbf{A} U \left|\psi\right\rangle = R\left\langle\psi\right| \mathbf{A} \left|\psi\right\rangle, \tag{4.19}$$

On the right-hand side, we make use of the Einstein summation convention to write

$$\Big[R\langle\mathbf{A}\rangle\Big]_i = R_{ij}\langle A_j\rangle = \langle R_{ij} A_j\rangle \tag{4.20}$$

Since the expression above holds for each component, it follows that

$$\left\langle\psi\right| U^\dagger \mathbf{A} U \left|\psi\right\rangle = \left\langle\psi\right| R\mathbf{A} \left|\psi\right\rangle \tag{4.21}$$

If A and B are Hermitian operators and $\langle\psi|A|\psi\rangle = \langle\psi|B|\psi\rangle$ for all states $|\psi\rangle$, then $A = B$ (you proved this statement in homework #2; it's not as obvious as it seems). We then arrive at the following adjoint formula for the vector operator $\mathbf{A}$:

$$\boxed{U^\dagger \mathbf{A} U = R\mathbf{A}} \tag{4.22}$$

The adjoint formula for rotations defines the relationship between a unitary rotation operator $U(R)$ and its corresponding rotation R. Notice the mapping is not one-to-one; the rotation R can be mapped to the unitary operators $U(R)$ and $-U(R)$. As it turns out, we've encountered an example of this equation during our study of spin precession for the vector operator $\mathbf{S}$. Let's investigate the adjoint formula in more detail.

Example 4.1: Rotations in Quantum Mechanics

Consider a spin-1/2 system. Verify Eq. (4.22) for rotation around the z axis by an angle φ by explicit computation. Use the matrix representations in the S_z basis.

Solution: According to Eq. (4.22),

$$U^\dagger(\hat{\mathbf{z}},\varphi)\mathbf{S}U(\hat{\mathbf{z}},\varphi) = R(\hat{\mathbf{z}},\varphi)\mathbf{S}$$

$$\begin{pmatrix} \exp(i\varphi S_z/\hbar)S_x\exp(-i\varphi S_z/\hbar) \\ \exp(i\varphi S_z/\hbar)S_y\exp(-i\varphi S_z/\hbar) \\ \exp(i\varphi S_z/\hbar)S_z\exp(-i\varphi S_z/\hbar) \end{pmatrix} = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix}$$

Since S_z commutes with any analytic function of itself, the z component is trivially satisfied. Let's check the x component by explicit calculation. On the left-hand side,

$$\exp(i\varphi S_z/\hbar)S_x\exp(-i\varphi S_z/\hbar) = \begin{pmatrix} e^{i\varphi/2} & 0 \\ 0 & e^{-i\varphi/2} \end{pmatrix} \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2} \end{pmatrix}$$

$$= \frac{\hbar}{2}\begin{pmatrix} 0 & e^{i\varphi} \\ e^{-i\varphi} & 0 \end{pmatrix}$$

And on the right-hand side,

$$S_x\cos\varphi - S_y\sin\varphi = \frac{\hbar}{2}\begin{pmatrix} 0 & \cos\varphi \\ \cos\varphi & 0 \end{pmatrix} - \frac{\hbar}{2}\begin{pmatrix} 0 & -i\sin\varphi \\ i\sin\varphi & 0 \end{pmatrix}$$

$$= \frac{\hbar}{2}\begin{pmatrix} 0 & e^{i\varphi} \\ e^{-i\varphi} & 0 \end{pmatrix}$$

As for the y component, work out the calculations for yourself and verify that both sides equal

$$\frac{\hbar}{2}\begin{pmatrix} 0 & -ie^{i\varphi} \\ ie^{-i\varphi} & 0 \end{pmatrix}$$

It's good to see that the expression works when we're given the observables for a physical system, but ultimately Eq. (4.22) is a foundational relationship that connects unitary rotation operators to their corresponding rotations in 3D Euclidean space. In the next section, we'll see how the observables corresponding to the components of $\mathbf{S}$ in a spin-1/2 system (and others) naturally arise from the quantum theory of rotations.

Example 4.2: Spin Precession

Let's revisit spin precession from chapter #3 in the context of quantum rotations. Consider a uniform magnetic field along the z axis such that the Hamiltonian takes the form $H = \omega_0 S_z$ and the unitary time evolution operator is

$$U(t) = \exp(-i\omega_0 S_z t/\hbar)$$

We recognize this as a unitary rotation operator related to rotations about the z axis, since S_z is the generator of rotations about the z axis. Given the initial state

$$|\psi(0)\rangle = \cos(\theta/2)\,|+z\rangle + e^{-i\phi}\sin(\theta/2)\,|-z\rangle\,,$$

we know the expectation value of $\mathbf{S}$ is

$$\langle \mathbf{S}\rangle = \frac{\hbar}{2}(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$$

From the adjoint formula, we can easily compute the time-evolved (rotated) vector:

$$\langle S_x(t)\rangle = \langle S_x\rangle\cos\omega_0 t - \langle S_y\rangle\sin\omega_0 t = \frac{\hbar}{2}\sin\theta\cos(\phi+\omega_0 t)$$

$$\langle S_y(t)\rangle = \langle S_x\rangle\sin\omega_0 t + \langle S_y\rangle\cos\omega_0 t = \frac{\hbar}{2}\sin\theta\sin(\phi+\omega_0 t)$$

and of course,

$$\langle S_z(t)\rangle = \langle S_z\rangle = \frac{\hbar}{2}\cos\theta$$

We have obtained the same results using the adjoint formula, as expected. Note, once again, that our results relied on having the correct spin component operators, so ultimately this result is circular. We determined the spin operators as far as we could from experimental considerations alone. If you remember, we had to adopt a few "phase conventions," and we can ultimately trace those to the theory of rotations in quantum mechanics, and in particular, the adjoint formula.

4.2.3 Commutation Relations

Consider a finite rotation $R_i(\theta) = R(\hat{\mathbf{e}}_i, \theta)$ about the i-th Cartesian coordinate axis, where $i = 1, 2, 3$ corresponds to x, y, z respectively. From the adjoint formula,

$$U_i^\dagger(\theta) A_j U_i(\theta) = R_i(\theta)_{jk} A_k, \tag{4.23}$$

where $U_i(\theta) = U(\hat{\mathbf{e}}_i, \theta)$ is the unitary operator associated with $R_i(\theta)$ and $R_i(\theta)_{jk}$ denotes the components of R_i. Note, once again, we're using the summation convention with an implied sum

over the repeated index k. From Eq. (4.23), it follows that

$$\frac{d}{d\theta}\Big[U_i^\dagger(\theta)A_jU_i(\theta)\Big]_{\theta=0} = \frac{d}{d\theta}\Big[R_i(\theta)_{jk}A_k\Big]_{\theta=0} \tag{4.24}$$

On the left-hand side,

$$\begin{aligned}\frac{d}{d\theta}\Big[U_i^\dagger(\theta)A_jU_i(\theta)\Big]_{\theta=0} &= \frac{d}{d\theta}\Big[U_i^\dagger(\theta)\Big]_{\theta=0}A_jU_i(0) + U_i^\dagger(0)A_j\frac{d}{d\theta}\Big[U_i(\theta)\Big]_{\theta=0} \\ &= \frac{i}{\hbar}J_iA_j - \frac{i}{\hbar}A_jJ_i \\ &= \frac{i}{\hbar}[J_i, A_j]\end{aligned}$$

And on the right-hand side,

$$\frac{d}{d\theta}\Big[R_i(\theta)_{jk}A_k\Big]_{\theta=0} = \frac{d}{d\theta}\Big[R_i(\theta)_{jk}\Big]_{\theta=0}A_k = \big[\mathcal{J}_i\big]_{jk}A_k$$

where $\big[\mathcal{J}_i\big]_{jk}$ denotes the components of the classical rotation generator $\mathcal{J}_i$. At this point, we introduce the three-dimensional Levi-Civita symbol given by

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \text{ is } (1,2,3),(3,1,2),(2,3,1) \\ -1 & \text{if } (i,j,k) \text{ is } (3,2,1),(2,1,3),(1,3,2) \\ 0 & \text{if any two indices are equal} \end{cases} \tag{4.25}$$

An easy way to remember the nonzero components of the Levi-Civita is to remember that it's $+1$ when (i,j,k) is an even permutation of (1,2,3), meaning the permutation is formed by an even number of transpositions, and it's -1 when (i,j,k) is an odd permutation, meaning the permutation is formed by an odd number of transpositions. In 3D only, the cyclic permutations of $(1,2,3)$ are all even permutations, and the anticyclic permutations are all odd permutations. Incidentally, $\big[\mathcal{J}_i\big]_{jk} = -\varepsilon_{ijk}$, as you can easily verify[1], and it follows that

$$\frac{d}{d\theta}\Big[R_i(\theta)_{jk}A_k\Big]_{\theta=0} = -\varepsilon_{ijk}A_k$$

Putting together the left- and right-hand sides, we find

$$[J_i, A_j] = i\hbar\varepsilon_{ijk}A_k \tag{4.26}$$

On its own, this result is noteworthy, but we are primarily interested in the case $\mathbf{A} = \mathbf{J}$. We finally arrive at the commutation relations for the angular momentum operators:

$$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k \tag{4.27}$$

[1]Note the convention here is that with i fixed, j indicates the row and k indicates the column.

where, as another reminder, we are using the summation convention. In particular,

$$[J_x, J_y] = i\hbar J_z, \quad [J_z, J_x] = i\hbar J_y, \quad [J_y, J_z] = i\hbar J_x \tag{4.28}$$

We have achieved something rather remarkable; starting only from the geometric properties of rotations in ordinary three-dimensional space and the requirement that quantum expectation values transform in the same way as classical vectors under these rotations, we have deduced—without ever referring to any specific physical system—that the generators of rotations in quantum mechanics must satisfy the commutation relations in Eq. (4.28).

The commutation relations of angular momentum not only encode the structure of rotations but also determine the allowed values, or spectrum, of angular momentum in quantum systems. Because the components of angular momentum do not commute, it is impossible to know all three simultaneously. Instead, we can diagonalize J^2 and one component, usually J_z, and we find discrete, quantized values for both. The commutation relations define a Lie algebra—specifically, the algebra of infinitesimal rotations—and it is the representation theory of this algebra that constrains the possible measurement outcomes and leads to the quantized spectrum of angular momentum.

4.3 Representations of Angular Momentum

We will now explore the consequences of the angular momentum commutation relations for the construction of angular momentum eigenstates. Consider the operator

$$J^2 = J_x^2 + J_y^2 + J_z^2 \tag{4.29}$$

and note that $[J^2, J_x] = [J^2, J_y] = [J^2, J_z] = 0$. Since J^2 commutes with the components of $\mathbf{J}$, we can construct simultaneous eigenstates of J^2 and any one of the components. However, since these components do not commute with each other, we cannot find simultaneous eigenstates of more than one component of $\mathbf{J}$. By convention we look for simultaneous eigenstates of J^2 and J_z. Every basis vector is labeled by two numbers (the eigenvalues)

$$J^2 \left|\alpha, \beta\right\rangle = \hbar^2 \alpha \left|\alpha, \beta\right\rangle \tag{4.30}$$

$$J_z \left|\alpha, \beta\right\rangle = \hbar \beta \left|\alpha, \beta\right\rangle \tag{4.31}$$

We'll assume without loss of generality that the eigenstates are orthonormal. Note that

$$J^2 - J_z^2 = J_x^2 + J_y^2 \geq 0$$

since J_x and J_y are self-adjoint. It follows that

$$\alpha \geq \beta^2$$

Since $0 \leq \alpha < \infty$, β is bounded above and below for a given value of α. In other words,

$$-\sqrt{\alpha} \leq \beta \leq \sqrt{\alpha}$$

Therefore, β has an upper bound $\beta_{\max}$ and a lower bound $\beta_{\min}$.

Whenever eigenvalues are bounded, it is useful to search for *raising* and *lowering* operators that can be used to construct eigenvectors by repeated application. The only generators that can map an eigenstate to a different vector are J_x and J_y, and after careful consideration, we define the following raising and lowering operators (ladder operators)

$$J_+ = J_x + iJ_y, \quad J_- = J_x - iJ_y \tag{4.32}$$

I realize this is the least motivated step in the whole process, but it really is the key that unlocks the eigenstates. This process of considering a complex linear combination, instead of the original generators, is called a *complexification*, and it's a common technique. The commutation relations for the ladder operators are

$$[J_z, J_\pm] = \pm\hbar J_\pm, \quad [J_+, J_-] = 2\hbar J_z \tag{4.33}$$

Notice that complexification breaks the cyclical symmetry of the original commutation relations. Not only that, but we can use the commutation relations to justify the claim that $J_\pm$ are the raising and lowering operators. For starters, $J_\pm$ does not alter the eigenvalue of J^2 since $[J^2, J_\pm] = 0$.

$$J^2 \left(J_\pm \left|\alpha, \beta\right\rangle\right) = J_\pm J^2 \left|\alpha, \beta\right\rangle = \hbar^2 \alpha \left(J_\pm \left|\alpha, \beta\right\rangle\right)$$

From the commutator of L_z and $L_\pm$, it follows that

$$\begin{aligned} J_z \left(J_\pm \left|\alpha, \beta\right\rangle\right) &= J_\pm J_z \left|\alpha, \beta\right\rangle + \left[J_z, J_\pm\right] \left|\alpha, \beta\right\rangle \\ &= \hbar\beta J_\pm \left|\alpha, \beta\right\rangle \pm \hbar J_\pm \left|\alpha, \beta\right\rangle \\ &= \hbar(\beta \pm 1) \left(J_\pm \left|\alpha, \beta\right\rangle\right) \end{aligned}$$

Hence, $J_\pm \left|\alpha, \beta\right\rangle$ is proportional to $\left|\alpha, \beta \pm 1\right\rangle$. We'll find the constant of proportionality later by requiring that $\left|\alpha, \beta\right\rangle$ is an orthonormal set. Since β is bounded above and below,

$$J_+ \left|\alpha, \beta_{\max}\right\rangle = 0 \quad \text{and} \quad J_- \left|\alpha, \beta_{\min}\right\rangle = 0 \tag{4.34}$$

Furthermore,

$$\begin{aligned} 0 = J_- J_+ \left|\alpha, \beta_{\max}\right\rangle &= \left(J_x^2 + J_y^2 - \hbar J_z\right) \left|\alpha, \beta_{\max}\right\rangle \\ &= \left(J^2 - J_z^2 - \hbar J_z\right) \left|\alpha, \beta_{\max}\right\rangle \\ &= \left(\hbar^2 \alpha - \hbar^2 \beta_{\max}^2 - \hbar^2 \beta_{\max}\right) \left|\alpha, \beta_{\max}\right\rangle \end{aligned}$$

$$\alpha = \beta_{\max} \left(\beta_{\max} + 1\right)$$

Similarly, using $J_- \left|\alpha, \beta_{\min}\right\rangle = 0$ and $J_+ J_- = J^2 - J_z^2 + \hbar J_z$, we find

$$\alpha = \beta_{\min} \left(\beta_{\min} - 1\right)$$

We now introduce the standard notation $j = \beta_{\max}$ and $\beta = m$ and relabel the eigenstates

$$J^2 \left|j, m\right\rangle = \hbar^2 j(j+1) \left|j, m\right\rangle \quad \text{and} \quad J_z \left|j, m\right\rangle = \hbar m \left|j, m\right\rangle$$

where

$$-j \leq m \leq j \tag{4.35}$$

Starting from the vector $|j, j\rangle$, we obtain the vector $|j, -j\rangle$ by repeated application of J_-. After each application, the eigenvalue is lowered by one, so the difference between the highest and lowest eigenvalue must be an integer n. It follows that

$$j - (-j) = 2j = n \quad \rightarrow \quad j = \frac{n}{2} \tag{4.36}$$

Hence, the only values of j allowed are non-negative integers or half-integers:

$$j = 0,\ \frac{1}{2},\ 1,\ \frac{3}{2},\ \ldots$$

When we say that j belongs to the indicated set of values (integers and half-integers), we mean that the commutation relations alone tell us that j can take on only these values. They do not tell us which of these values actually occur in a specific application.

Let's look at the eigenvalues for the first few representations:

$$\begin{array}{lll} j = 0 & j(j+1) = 0 & m = 0 \\ j = 1/2 & j(j+1) = 3/4 & m = -1/2, 1/2 \\ j = 1 & j(j+1) = 2 & m = -1, 0, 1 \\ \ldots & & \end{array}$$

The dimension of the vector space is $2j + 1$ for a given representation, and we can construct the

eigenvectors of J_z using the ladder operators (the raising the lowering operators). Recall

$$J_\pm |j,m\rangle = C_\pm |j,m\pm 1\rangle$$

for constants $C_\pm$ that depend on j and m. The norm squared of $J_+ |j,m\rangle$ is

$$\begin{aligned}\|J_+ |j,m\rangle\|^2 = |C_+|^2 &= \langle j,m| J_+^\dagger J_+ |j,m\rangle \\ &= \langle j,m| J_- J_+ |j,m\rangle \\ &= \langle j,m| \left(J^2 - J_z^2 - \hbar J_z\right) |j,m\rangle \\ &= \hbar^2 \left(j(j+1) - m^2 - m\right)\end{aligned}$$

A similar calculation gives $|C_-|^2$. There is ambiguity in the overall phase factor, so we choose

$$C_\pm = \hbar\sqrt{j(j+1) - m(m\pm 1)} \tag{4.37}$$

Given an eigenstate $|j,m\rangle$, a ladder operator will produce a higher or lower eigenstate:

$$J_\pm |j,m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\, |j,m\pm 1\rangle \tag{4.38}$$

4.3.1 Spin 1/2 and SU(2)

The smallest non-trivial value of j is $1/2$. The generator J_z has eigenvalues $\hbar/2$ and $-\hbar/2$, and because J_z is conventionally chosen to be diagonal,

$$J_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The corresponding eigenvectors are

$$\left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \left|\tfrac{1}{2},-\tfrac{1}{2}\right\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{4.39}$$

As for the ladder operators,

$$J_+ \left|\tfrac{1}{2},-\tfrac{1}{2}\right\rangle = \hbar \left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle \quad \text{and} \quad J_- \left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle = \hbar \left|\tfrac{1}{2},-\tfrac{1}{2}\right\rangle$$

and their corresponding matrix representations are

$$J_+ = \hbar\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad J_- = \hbar\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \tag{4.40}$$

From $J_+ = J_x + iJ_y$ and $J_- = J_x - iJ_y$, we can invert the expressions to find

$$J_x = \frac{1}{2}(J_+ + J_-) = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$J_y = \frac{1}{2i}(J_+ - J_-) = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

These are the spin-1/2 angular momentum operators we inferred from experiments earlier in the course! Keep in mind there is no guarantee that any physical system will align with this mathematical description. As it turns out, if you combine quantum mechanics, special relativity, and rotational invariance, *spin is unavoidable*. You would learn more about this in a course on quantum field theory.

There's a topic I've been putting off for a while, but now seems like a good time to discuss the group SU(2), the group of special unitary 2×2 complex matrices. In the same way that complex numbers can be used to represent 2D rotations (rotation around a single, fixed axis), there is a way to use 2×2 special unitary matrices to represent rotations in 3D Euclidean space. I'd like to show you how that works.

The unitary rotation operators in a spin-1/2 system belong to SU(2). They are 2×2 complex, unitary matrices, and they are "special" because $\det U = 1$. In the axis-angle parameterization, any unitary rotation operator in a spin-1/2 system can be written as

$$U(\hat{\mathbf{n}}, \theta) = \exp\left(-\frac{i}{\hbar}\theta\hat{\mathbf{n}} \cdot \mathbf{S}\right) = \exp\left(-\frac{i}{2}\theta\hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right)$$

where we use $\mathbf{S}$ to denote angular momentum, and $\boldsymbol{\sigma}$ denotes the **Pauli matrices**:

$$\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z) \quad \text{where} \quad S_x = \frac{\hbar}{2}\sigma_x,$$

and for reference,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Any 2×2 Hermitian matrix can be written as a linear combination of the identity operator and the Pauli matrices. To see why, consider an arbitrary 2×2 Hermitian matrix

$$A = \begin{pmatrix} a_0 + a_z & a_x + ia_y \\ a_x - ia_y & a_0 - a_z \end{pmatrix}$$

where c and the components of $\mathbf{a} = (a_x, a_y, a_z)$ are real numbers. It follows that

$$A = a_0 \mathbb{1} + \mathbf{a} \cdot \boldsymbol{\sigma}$$

This result suggests that any 2×2 Hermitian matrix encodes four parameters, and three of them correspond to a vector in 3D Euclidean space.

Just as in the Heisenberg picture, we regard state vectors to be stationary while the operators evolve unitarily over time, we can do the same thing for rotations. Start by considering the expectation value of A under unitary rotation of the state $|\psi\rangle$:

$$\langle \psi' | A | \psi' \rangle = \langle \psi | U^\dagger A U | \psi \rangle$$

The unitary transformation of A is then

$$A' = U^\dagger A U$$

which has the same structure as a time-evolving Heisenberg operator. It follows that

$$A' = U^\dagger (a_0 \mathbb{1} + \mathbf{a} \cdot \boldsymbol{\sigma}) U = a_0 \mathbb{1} + U^\dagger (\mathbf{a} \cdot \boldsymbol{\sigma}) U$$

Using Eq. (4.22), we can expand the second term as

$$U^\dagger (\mathbf{a} \cdot \boldsymbol{\sigma}) U = \sum_i a_i U^\dagger \sigma_i U = \sum_i a_i \sum_j R_{ij} \sigma_j$$

Note that $R_{ji} = R^T_{ji}$ such that

$$U^\dagger (\mathbf{a} \cdot \boldsymbol{\sigma}) U = \sum_j \left(\sum_i R^T_{ji} a_i \right) \sigma_j = \left(R^T \mathbf{a} \right) \cdot \boldsymbol{\sigma}$$

Altogether, we find

$$A' = U^\dagger A U = a_0 \mathbb{1} + \left(R^T \mathbf{a} \right) \cdot \boldsymbol{\sigma}$$

In the *Schrödinger picture of rotations*, the state vectors unitarily rotate in correspondence with the classical rotation R of a Bloch vector. In the *Heisenberg picture of rotations*, state vectors are fixed and the operators are unitarily rotating in the opposite sense since $R^T = R^{-1}$. This is certainly an interesting result, but let's revisit that second term. Let $R^T \to R$ and note that $U(R^T) = U^\dagger(R)$. Hence,

$$U(R) (\mathbf{a} \cdot \boldsymbol{\sigma}) U^\dagger(R) = (R\mathbf{a}) \cdot \boldsymbol{\sigma}$$

This result, also known as an adjoint formula for rotations, constitutes a mapping from elements of SU(2) to elements of SO(3). Notice the mapping is two-to-one because both $U(R)$ and $-U(R)$ map to the same rotation matrix R.

4.3.2 Spin 1

In the case $j = 1$,

$$J_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad J_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{4.41}$$

and the basis vectors are

$$|1,1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1,0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1,-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{4.42}$$

Example 4.3: Stern-Gerlach experiment with spin-1 particles

Consider a Stern-Gerlach experiment with a neutral beam of spin-1 particles. What fraction of particles with $S_y = \hbar$ will be found to have $S_z = \hbar$ with the appropriate arrangement of analyzers?

Solution: The quantitative answer to this equation is the square of an inner product:

$$|\langle 1,1|1,1\rangle_y|^2$$

where $|1,1\rangle$ is the eigenvector of S_z with eigenvalue $\hbar$ and $|1,1\rangle_y$ is the eigenvector of S_y with eigenvalue $\hbar$. Find $|1,1\rangle_y$ in the basis of S_z. The eigenvalue equation is

$$\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \hbar \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

From which it follows that

$$-ib = \sqrt{2}a, \quad i(a-c) = \sqrt{2}b, \quad ib = \sqrt{2}c$$

Hence, $b = i\sqrt{2}a$ and $c = -a$. After normalizing the state,

$$|1,1\rangle_y = \frac{1}{2} \begin{pmatrix} 1 \\ i\sqrt{2} \\ 1 \end{pmatrix}$$

and finally,

$$|\langle 1,1|1,1\rangle_y|^2 = \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4}$$

Furthermore, 1/2 of the particles will have $S_z = 0$ and 1/4 of them will have $S_z = -\hbar$. **Your turn.** What fraction of particles with $S_y = 0$ with be found to have $S_z = \hbar, 0$ and $-\hbar$ with the appropriate arrangement of analyzers?